Lightweight Tomato Leaf Disease Detection via YOLO26 with MobileNetV4 Backbone and SimAM Attention

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***Abstract*—To solve the problems of large lesion scale difference, relatively similar textures and high calculation cost in tomato leaf disease detection, this paper proposes a lightweight detection method based on YOLO26, which integrates a lightweight backbone network and parameter-free attention mechanism. Firstly, the YOLOv8, YOLOv12 and YOLO26 were compared under a unified training configuration. YOLO26 was selected as the basis for subsequent improvements because it achieved better mAP@0.5:0.95 metrics on the target dataset. Then introduce MobileNetV4 to replace the original backbone network, reducing the number of parameters and the number of floating-point operations.When performing feature enhancement, compare SimAM, ACA, and BiFPN. SimAM was chosen because it improves detection accuracy while keeping the total model parameters and GFLOPs unchanged. Experiments were carried out on the public YOLO-format tomato leaf disease dataset, which has 10 categories and 10,853 images. The results show that the final YOLO26-MobileNetV4-SimAM reduces the parameters from 2.57M to 2.14M, and the computational complexity decreases from 6.1 GFLOPs to 4.4GFLOPs, mAP@0.5 rises from 0.97438 to 0.97604, mAP@0.5:0.95 increases from 0.83123 to 0.85909.The proposed method has competitive detection performance and reduces model complexity, and provides a reference for designing lightweight tomato leaf disease detection models.**

Keywords—tomato leaf disease detection; YOLO26; lightweight detection; MobileNetV4; SimAM; object detection

I. Introduction

Tomato is an important economic crop. Leaf diseases will affect plant growth, fruit quality and yield. With the development of computer vision and deep learning, plant disease and insect pest recognition has changed from manual diagnosis to automatic image analysis, and there are tasks such as classification, detection, segmentation [1].

Tomato leaf disease detection will encounter problems such as illumination variation, leaf occlusion, small lesions, blurred boundaries, and high inter-class texture similarity. Previous studies proposed that adding attention mechanisms to YOLO-based frameworks, performing multi-scale feature fusion, or adopting lightweight structures can improve tomato leaf disease detection in complex backgrounds [2]. Recent studies that improve YOLOv8 and YOLOv9 for tomato disease detection also often take accuracy, number of parameters, GFLOPs, and edge applications as the key points of evaluation [3,4]. However, reducing the model size while ensuring detection performance still needs to be improved in the design of agricultural visual inspection models.

The YOLO series adopts one-stage detection method, with relatively compact structure, direct inference and high efficiency[5]. YOLOv8 is widely used in agriculture, transportation, industrial detection and other fields, YOLOv12 emphasizes the attention mechanism in real-time detection[6]. This paper compares YOLOv8, YOLOv12 and YOLO26 on the same tomato leaf disease data, and then uses YOLO26 for lightweight improvement.

The contributions of this paper are: 1) Establish baselines for YOLOv8, YOLOv12, and YOLO26 under unified training; 2) Use MobileNetV3, MobileNetV4, and RepViT as backbones for comparison, and select MobileNetV4 for the lightweight version; 3) Use SimAM, ACA, and BiFPN as feature modules for comparison, and build the final model after ablation studies. By analyzing mAP, number of parameters, and GFLOPs, ensure that the model evaluation conforms to actual situations.

II. Dataset And Experimental Settings

A. Dataset

Using the Tomato Leaf Disease Detection - YOLOv8 dataset publicly available on Kaggle as experimental data. This dataset is used for tomato leaf disease object detection, containing 10,853 images with YOLO-format bounding box annotations. The categories are 10 tomato leaf conditions/diseases: Tomato Bacterial Spot, Tomato Early blight, Tomato healthy, Tomato Late blight, Tomato Leaf Mold, Tomato mosaic virus, Tomato Septoria leaf spot, Tomato Spider mites Two-spotted spider mite, Tomato Target Spot, Tomato Yellow Leaf Curl Virus[7].

B. Experimental Settings

To ensure the comparability of the models, all models are trained/evaluated using the same data split and the same input size. The input image size is 640×640, the training epoch is 80, and the batch size is 16. Automatic mixed precision is enabled during training to reduce memory and improve efficiency. YOLO26 and its improved variants use the MuSGD optimizer, and other models use the SGD optimizer.

The evaluation metrics include mAP@0.5, and also mAP@0.5:0.95, the number of parameters, and GFLOPs. Here, mAP@0.5 refers to the case where the mean average precision is when the intersection over union is 0.5; mAP@0.5:0.95 is

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the average detection performance under multiple intersection over union ratios, which can better reflect the positioning quality. The number of parameters and floating-point operations are indicators for calculating model size and computational complexity, and are important standards for comparing improvements of different structures.The relevant definitions are in Equations (1) through (3).

Quantitative performance detection is performed on the evaluation indicators. In Equation (1), TP, FP, and FN refer to true positive, false positive, and false negative respectively. In Equation (2), B_p is the predicted box and B_g is the ground truth box. In Equation (3), mAP_t is the mean average precision at IoU threshold t.

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN} \tag{1}$$

$$\text{IoU} = \frac{|B_p \cap B_g|}{|B_p \cup B_g|} \tag{2}$$

$$\text{mAP}_{0.5:0.95} = \frac{1}{10}\sum_{i=0}^{9} \text{mAP}_{0.50+0.05i} \tag{3}$$

III. Model Comparison And Improvement Method

A. Baseline Detection Model Comparison

Looking for the baseline model for subsequent improvements, first compare YOLOv8, YOLOv12 and YOLO26. YOLOv8 is a commonly used open-source detection framework with mature training and inference processes, YOLOv12 is a real-time detection framework focused on attention[6]. Table 1 shows that YOLOv8 and YOLO26 are relatively similar in terms of mAP@0.5 value, but YOLO26 achieved a value of 0.83123 on mAP@0.5:0.95 is higher than YOLOv8's 0.81091. This shows that under a more strict intersection over union, YOLO26 has better overall positioning. The mAP of YOLOv12 is lower than both YOLOv8 and YOLO26, so it is not selected for subsequent improvements.

Table 1. Comparison results of baseline detection models.

Model	mAP@0.5	mAP@0.5:0.95
YOLOv8	0.97426	0.81091
YOLOv12	0.88558	0.71684
YOLO26	0.97438	0.83123

B. Comparison of Lightweight Backbone Networks

After selecting YOLO26 as the baseline, compare different backbone replacement schemes. MobileNetV3 improves the efficiency of mobile models through neural architecture search and lightweight model design [8]. MobileNetV4 develops a general and efficient architecture suitable for the mobile ecosystem [9]. RepViT redesigns mobile CNNs from the perspective of lightweight Vision Transformers [10]. This paper introduces these three structures as YOLO26 backbone replacements, and compares them in terms of parameter count, GFLOPs, and mAP.

To explain the lightweight backbone replacement theory, compare the computational cost of standard convolution and depthwise separable convolution. For an input feature map of H×W, C_in is the number of input channels, C_out is the number of output channels, and the kernel size is k, the theoretical computational cost is in Equations (4) and (5), and the relative computational ratio is in Equation (6).

$$C_{\text{std}} = HWC_{\text{in}}C_{\text{out}}k^2 \tag{4}$$

$$C_{\text{ds}} = HWC_{\text{in}}k^2 + HWC_{\text{in}}C_{\text{out}} \tag{5}$$

$$R = \frac{C_{\text{ds}}}{C_{\text{std}}} = \frac{1}{C_{\text{out}}} + \frac{1}{k^2} \tag{6}$$

Equation (6) shows that large output channels and fixed kernel size make depthwise separable convolution greatly reduce the theoretical calculation cost, which meets the design goal of lightweight feature extraction of MobileNet series networks.

In Table 2, use MobileNetV4 to replace the original backbone network, the parameters are reduced from 2.57M to 2.14M, and GFLOPs are reduced from 6.1 to 4.4.mAP@0.5:0.95 changes from 0.83123 to 0.84953.MobileNetV3 has low computational cost for parameters, but it is inferior to MobileNetV4 in terms of mAP@0.5:0.95. RepViT has high detection accuracy, but its parameters and GFLOPs increase significantly, which does not meet the demand for reducing model computational load. So MobileNetV4 was chosen as the backbone network for the final combined model.

Table 2. Comparison results of different YOLO26 backbone networks.

Model Variant	Backbone	Params/M	GFLOPs	mAP@0.5	mAP@0.5:0.95
YOLO26-Baseline	Original	2.57	6.1	0.97438	0.83123
YOLO26-MobileNetV4	MobileNetV4	2.14	4.4	0.97253	0.84953
YOLO26-MobileNetV3	MobileNetV3	2.21	4.6	0.97164	0.84130
YOLO26-RepViT	RepViT	6.45	17.2	0.98175	0.87012

C. Comparison of Feature Enhancement Modules

In feature enhancement, SimAM, ACA and BiFPN are compared. SimAM uses an energy function to estimate the importance of neurons in the 3D feature map, and generates attention weights when there are no learnable parameters[11]. The ACA module is a candidate attention structure designed based on the frequency-spatial fusion network asymmetric cross-domain attention [12]. BiFPN is a weighted bidirectional feature pyramid structure from EfficientDet for multi-scale feature fusion [13]. To keep the analysis consistent with the experimental evidence, the paper only analyzes the obtained parameter count, GFLOPs and mAP results.

The core of SimAM is to estimate the importance of features, by relying on minimizing the energy of neurons. The

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closed-form solution can be simplified into Equation (7), in which t is the current neuron, μ and σ² are the mean and variance of neurons in the same channel, and λ is the stabilization term.

$$e_t^* = \frac{4(\sigma^2 + \lambda)}{(t-\mu)^2 + 2\sigma^2 + 2\lambda}$$

$$\tilde{X} = \text{sigmoid}\left(\frac{1}{E}\right) \odot X \tag{7}$$

Use the attention weights in this expression to recalibrate the feature responses, so that SimAM becomes a feature enhancement module without additional learnable parameters.

In Table 3, after introducing SimAM, the parameter count of YOLO26 is 2.57M, and GFLOPs is 6.1, there is no new parameter or GFLOPs change in the summary. mAP@0.5 and mAP@0.5:0.95 reach 0.98433 and 0.86186. ACA gets the same mAP@0.5:0.95 of 0.86186, the number of parameters and the number of gigaflops are higher than the baseline. BiFPN reduces the number of parameters, but its mAP@0.5:0.95 is smaller than SimAM. SimAM is selected as the final feature enhancement module, in terms of detection accuracy and module complexity.

TABLE 3. COMPARISON RESULTS OF DIFFERENT FEATURE ENHANCEMENT MODULES.

Model Variant	Module	Params/M	GFLOPs	mAP@0.5	mAP@0.5:0.95
YOLO26-Baseline	None	2.57	6.1	0.97438	0.83123
YOLO26-SimAM	SimAM	2.57	6.1	0.98433	0.86186
YOLO26-ACA	ACA	2.71	6.4	0.98138	0.86186
YOLO26-BiFPN	BiFPN	1.81	5.7	0.97958	0.85503

D. Ablation Study

To further analyze the impact of MobileNetV4 and SimAM on model performance, ablation experiments were conducted to compare the YOLO26 baseline, YOLO26 with MobileNetV4, YOLO26 with SimAM, and the final combined model, the results are shown in Table 4. Introducing SimAM alone can achieve the highest detection accuracy, but there is no reduction in model size. Using MobileNetV4 alone greatly reduces model complexity and improves mAP@0.5:0.95 compared with the baseline. The final combined model has 2.14M parameters, 4.4 GFLOPs, and mAP@0.5 is 0.97604 and mAP@0.5:0.95 is 0.85909, higher than the YOLO26 baseline, close to the case that only includes the SimAM variant.

TABLE 4. ABLATION RESULTS OF YOLO26 IMPROVEMENT SCHEMES.

Model	MobileNetV4	SimAM	Params/M	GFLOPs	mAP@0.5	mAP@0.5:0.95
YOLO26-Baseline	×	×	2.57	6.1	0.97438	0.83123
YOLO26-MobileNetV4	√	×	2.14	4.4	0.97253	0.84953
YOLO26-SimAM	×	√	2.57	6.1	0.98433	0.86186
YOLO26-MobileNetV4-SimAM	√	√	2.14	4.4	0.97604	0.85909

IV. RESULTS AND DISCUSSION

YOLO26 - MobileNetV4 - SimAM model combines MobileNetV4 backbone network and SimAM attention module. Compared with the YOLO26 basic model, the parameters of this model are reduced from 2.57M to 2.14M, calculated according to Equation (8), with a reduction of about 16.7%.GFLOPs changes from 6.1 to 4.4, a decrease of about 27.9%. Object detection accuracy, mAP@0.5 changes from 0.97438 to 0.97604, mAP@0.5:0.95 from 0.83123 becomes 0.85909, an increase of about 2.79 percentage points.

Equation (8) is used to calculate the relative reduction. P_b and G_b are the baseline parameters and floating-point operations; P_m and G_m are the improved parameters and floating-point operations.

$$\eta_P = \frac{P_b - P_m}{P_b} \times 100\%$$

$$\eta_G = \frac{G_b - G_m}{G_b} \times 100\% \tag{8}$$

Experimental results: MobileNetV4 model reduces model parameters and computational complexity, SimAM improves detection metrics by relying on feature representation. It should be noted that: SimAM has the highest mAP, but there is no parameter and GFLOPs reduction. So the significance of the final combined model does not lie in pursuing the highest accuracy, but in maintaining or improving the main detection metrics when greatly reducing the computational complexity of the model. Therefore, the final model is a lightweight detection method, not a purely high-accuracy model.

mAP@0.5:0.95 has improved. The model shows better performance under multiple IoU thresholds, and the overall matching between prediction boxes and ground truth annotations is better. Combined with the ablation results, MobileNetV4 reduces the model size while retaining effective feature extraction, and SimAM further improves the detection performance of lightweight models. The final model achieves a relatively balanced state between accuracy and computational cost.

In terms of model complexity, the final model has fewer parameters and GFLOPs compared to the YOLO26 baseline, but mAP@0.5 and mAP@0.5:0.95 have improved. The combination of MobileNetV4 and SimAM improves the efficiency-accuracy trade-off of YOLO26 in tomato leaf disease detection, laying a foundation for subsequent engineering verification.

A. Training Process and Visual Analysis of the Final Model

To show the convergence and detection performance of the YOLO26 - MobileNetV4 - SimAM model, we visualize the training curves of Figure 1 to Figure 3, the F1 - confidence curves and the precision - recall curves. These figures provide additional evidence from the training stability, the confidence threshold selection, the category detection performance.

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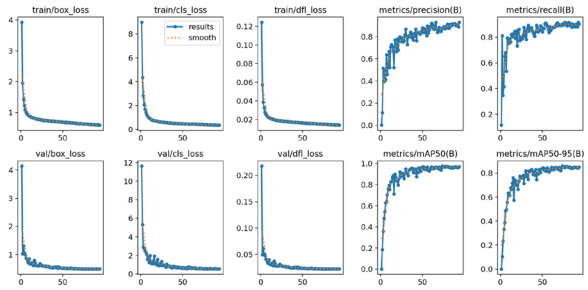


Fig. 1. Training losses and evaluation metrics of the final model.

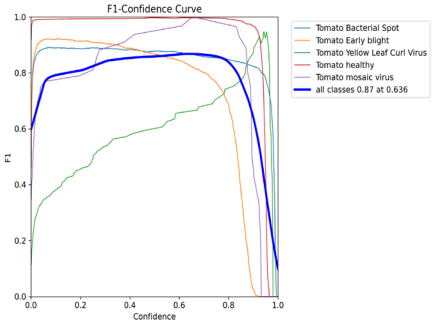


Fig. 2. F1-Confidence curve of the final model.

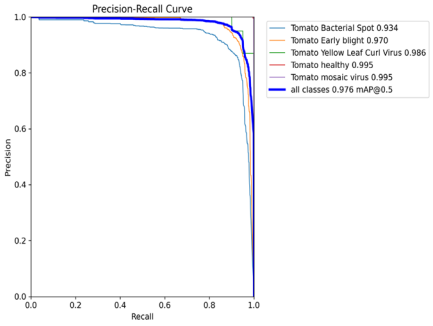


Fig. 3. Precision-Recall curve of the final model.

As shown in Figure 1, the box, class, and distribution focal losses of the training and validation stages decrease rapidly in the early stage and then stabilize. Precision, Recall, mAP@0.5 and mAP@0.5:0.95 generally show an upward trend, indicating that the 80-epoch training stably converges.In Figure 2, when the confidence threshold is approximately 0.636, the overall F1 score of each category is approximately 0.87, which is helpful for the selection of thresholds in the inference process. In Figure 3, the overall average mean precision at @0.5 is approximately 0.976 and several categories maintained high accuracy in the large-scale recall.

V. Conclusion

This paper focus on lightweight tomato leaf disease detection, and constructs the YOLO26 - MobileNetV4 - SimAM model based on YOLO26. This research does not directly stack modules, but follows the steps of baseline model selection, lightweight backbone replacement, feature enhancement module screening, and combined ablation verification. This chain provides a clear basis for the final structure. Regarding baseline comparison, on mAP@0.5:0.95, YOLO26 performs better than YOLOv8 and YOLOv12, which makes it a better basis for subsequent work to carry out improvements.

In lightweight design, MobileNetV4 replaces the original backbone network, the parameters are reduced from 2.57M to 2.14M, the GFLOPs are reduced from 6.1 to 4.4, and the mAP@0.5:0.95 is also improved from 0.83123 to 0.84953, backbone replacement makes the model smaller and still retains effective feature representations.When enhancing features, SimAM does not increase model parameters or GFLOPs within the model summary, and upwardly improves the mAP@0.5:0.95 of YOLO26 to 0.86186, reflecting the positive effect of parameter-free attention mechanisms on detection performance.

The ablation results show that, compared with the YOLO26 base model, the parameters of the final YOLO26 - MobileNetV4 - SimAM model are reduced by about 16.7%, GFLOPs are reduced by about 27.9%, and the mAP@0.5 rises from 0.97438 to 0.97604, and mAP@0.5:0.95 rises from 0.83123 to 0.85909. Combining the training curves andPR curves , the final model shows a state of stable convergence and good efficiency-accuracy balance on the tomato leaf disease detection data.

Overall, the proposed method has a simple structure and has been verified through experiments, with a good balance between model complexity and detection performance. Future work may extend the evaluation to more complex scenarios, category-level error analysis and specific hardware platforms, so as to improve the applicability and persuasiveness of the model in agricultural visual detection tasks.

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